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Application No: IN26012230

RAG Chatbot

Objective

The aim of this project is to build a chatbot that can answer questions based on the contents of a PDF document instead of searching the internet.

Tools Used

- Python 3.14
- Streamlit
- PyMuPDF
- Sentence Transformers (all-MiniLM-L6-v2)
- FAISS
- Ollama
- Qwen 3 8B

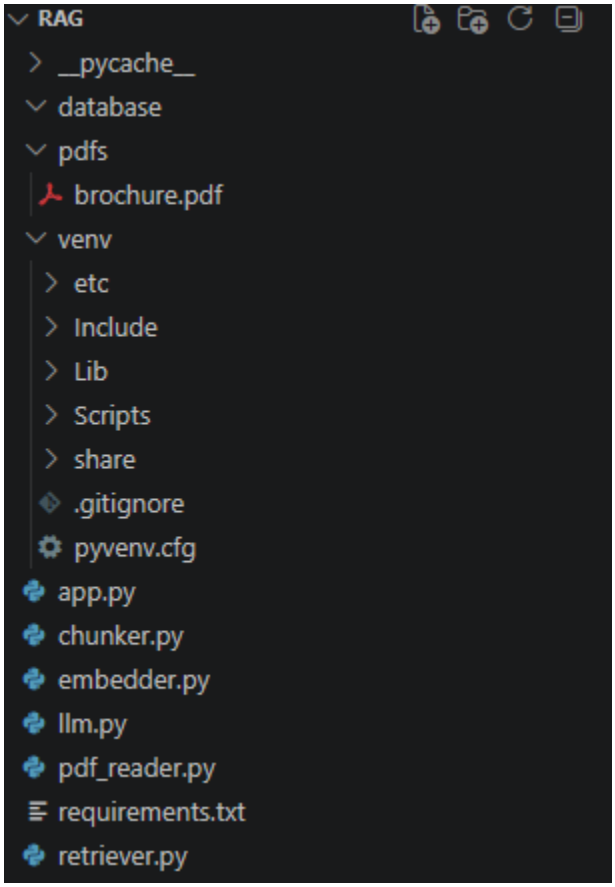
How It Works

1. 1. The user uploads a PDF document.
2. 2. The PDF is read and converted into text.
3. The text is divided into smaller chunks.
4. Each chunk is converted into embeddings.
5. The embeddings are stored in FAISS.

```
(venv) PS Z:\Projects\RAG> python pdf_reader.py
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|| 103/103 [00:00<00:00, 12639.36it/s]
Batches: 100%|| 13/13 [00:01<00:00,  8.96it/s]
Pages      : 52
Chunks     : 398
Embeddings : 398
Vector size: 384

FAISS index built!
Vectors stored: 398
```

6. Qwen 3 8B generates an answer using only the retrieved information.
7. The chatbot displays the source page numbers used for the answer Qwen 3 generates an answer using only the retrieved information.



Test Document

A Guide to Yale College 2024-2025.pdf (brochure pdf)

- Total Pages: 52
- Total Chunks: 398

Or Any of your choice as well

Sample Questions

Question	Result
What is the cost of attendance?	Correct answer generated
What financial aid is available?	Correct answer generated
What housing options are available?	Correct answer generated
How do I apply?	Correct answer generated
Who won the FIFA World Cup 2022?	Correctly responded that the information was not available in the document.

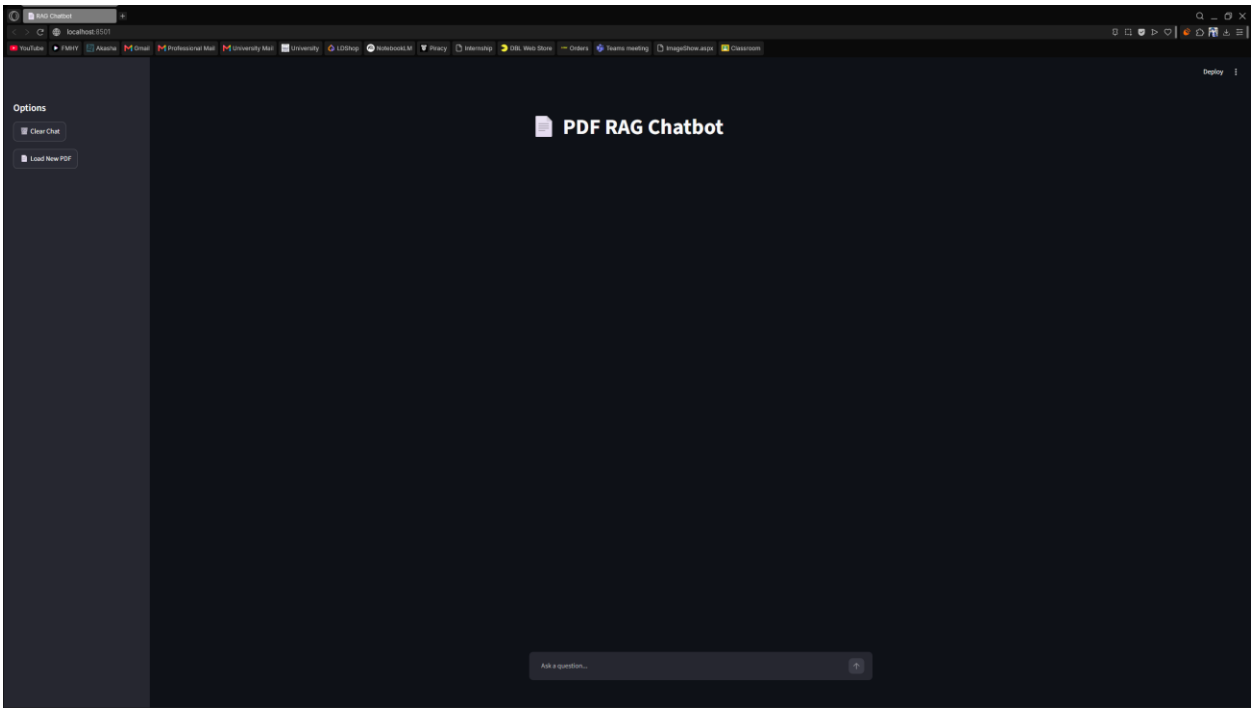
```
(venv) PS Z:\Projects\RAG> streamlit run app.py

  You can now view your Streamlit app in your browser.

  Local URL: http://localhost:8501
  Network URL: http://172.16.0.2:8501

  Help agents write better Streamlit apps?
  Install the official Streamlit skills by running streamlit skills in your terminal.

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|| 103/103 [00:00<00:00, 10911.36it/s]
Batches: 100%|| 13/13 [00:01<00:00,  7.35it/s]
Search took: 0.012 seconds
LLM took: 3.447 seconds
Loading weights: 100%|| 103/103 [00:00<00:00, 11113.17it/s]
Search took: 0.012 seconds
LLM took: 0.470 seconds
Search took: 0.014 seconds
LLM took: 2.458 seconds
Search took: 0.013 seconds
LLM took: 2.166 seconds
```





PDF RAG Chatbot

requirements to enter

The requirements to enter Yale College can be found on their admissions website at admissions.yale.edu, where applicants can find all admissions requirements.

Source Pages: 1, 17, 49

whats the total cost of living

The total cost of living for the 2024–2025 academic year at Yale is 87,150, which includes tuition & fees (67,250), room $(11, 300)$, $and board$(8,600).

Source Pages: 50

i can't afford the cost of living

Yale College admits students on the basis of academic and personal promise and without regard to their ability to pay. All aid is need-based. Once a student is admitted, Yale will meet 100% of that student's demonstrated financial need. This policy applies to all students, regardless of citizenship or immigration status.

Source Pages: 29, 38, 50

but i want to go to moon with elon musk

I couldn't find that information in the document.

Ask a question...

Conclusion

The chatbot successfully answered questions using information from the uploaded PDF. It also correctly rejected questions whose answers were not present in the document, demonstrating document-based Retrieval-Augmented Generation (RAG).